Parallel & Distributed Computing

LAB-3

Multi-Node Docker Swarm (Overlay Networking)

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2K23/BLCS/38

Manager Node Commands :

```
haroon@haroon-pc: ~

haroon@haroon-pc:~$ docker swarm init --advertise-addr 192.168.43.177
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
haroon@haroon-pc:~$ docker swarm leave --force
Node left the swarm.
haroon@haroon-pc:~$ docker swarm init --advertise-addr 192.168.43.177
Swarm initialized: current node (oenjz1lpyebmpn3lb5pq4nwaz) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-37jlu86ka5cmyvfaofzz85rwpc7h66wd9fsgd892kut5obpt25-eo5k8zz0ws0p42tvxj7zngnd4 192.168.43.177:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
haroon@haroon-pc:~$ docker node ls
ID                            HOSTNAME       STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
i0x95xsigalwkhwn6ucrmmaec     Time-Machine   Ready     Active                          26.1.5+dfsg1
oenjz1lpyebmpn3lb5pq4nwaz *   haroon-pc      Ready     Active         Leader           28.2.2
haroon@haroon-pc:~$ ping 192.168.43.160
PING 192.168.43.160 (192.168.43.160) 56(84) bytes of data.
64 bytes from 192.168.43.160: icmp_seq=1 ttl=128 time=11.5 ms
64 bytes from 192.168.43.160: icmp_seq=2 ttl=128 time=9.04 ms
64 bytes from 192.168.43.160: icmp_seq=3 ttl=128 time=10.9 ms
64 bytes from 192.168.43.160: icmp_seq=4 ttl=128 time=11.5 ms
64 bytes from 192.168.43.160: icmp_seq=5 ttl=128 time=11.8 ms
^C
--- 192.168.43.160 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 9.035/10.951/11.802/1.004 ms
haroon@haroon-pc:~$ docker service create \
--name nginxweb \
-p 8081:80 \
nginx
qt4yocru265g28dgoc7pe98zu
overall progress: 1 out of 1 tasks
1/1: running
verify: Service qt4yocru265g28dgoc7pe98zu converged
haroon@haroon-pc:~$ docker service scale nginxweb=2
nginxweb scaled to 2
overall progress: 2 out of 2 tasks
1/2: running
2/2: running
verify: Service nginxweb converged
```

```
haroon@haroon-pc: ~
haroon@haroon-pc:~$ docker service ps nginxweb
ID             NAME         IMAGE          NODE           DESIRED STATE   CURRENT STATE                ERROR     PORTS
xy96x3vgky0u   nginxweb.1   nginx:latest   haroon-pc      Running         Running about a minute ago
k2cj77fg4hc2   nginxweb.2   nginx:latest   Time-Machine   Running         Running 17 seconds ago
haroon@haroon-pc:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS     NAMES
fadc108b9a4b   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   80/tcp    nginxweb.1.xy96x3vgky0un97dbzo7ndw7v
haroon@haroon-pc:~$ ^[[200~docker run -d \
> -p 8080:8080 \
> -v /var/run/docker.sock:/var/run/docker.sock \
> dockersamples/visualizer~
docker: command not found
haroon@haroon-pc:~$ docker run -d \
-p 8080:8080 \
-v /var/run/docker.sock:/var/run/docker.sock \
dockersamples/visualizer
3c5884a3ca0dbb62a231e2b0924ad2e0fd58427a73a2c3a115dbefa0254f6d4f
haroon@haroon-pc:~$ docker node update --availability drain haroon-pc
haroon-pc
haroon@haroon-pc:~$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS                   PORTS                                         NAMES
3c5884a3ca0d   dockersamples/visualizer   "/sbin/tini -- node …"   2 minutes ago   Up 2 minutes (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   gallant_snyder
haroon@haroon-pc:~$ docker service rm nginxweb
nginxweb
haroon@haroon-pc:~$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS                   PORTS                                         NAMES
3c5884a3ca0d   dockersamples/visualizer   "/sbin/tini -- node …"   5 minutes ago   Up 5 minutes (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   gallant_snyder
haroon@haroon-pc:~$ ^[[200~docker service create \
> --name bda-mysql \
> -p 3306:3306 \
> -e MYSQL_ROOT_PASSWORD=123 \
> -e MYSQL_DATABASE=bda_db \
>
docker: command not found
haroon@haroon-pc:~$ docker service create \
--name bda-mysql \
-p 3306:3306 \
-e MYSQL_ROOT_PASSWORD=123 \
-e MYSQL_DATABASE=bda_db \
mysql
8nt70z5mxdcgbwqr2vpmwkrgs
```

```
haroon@haroon-pc: ~
8nt70z5mxdcgbwqr2vpmwkrgs
overall progress: 1 out of 1 tasks
1/1: running
verify: Service 8nt70z5mxdcgbwqr2vpmwkrgs converged
haroon@haroon-pc:~$ docker service ls
ID             NAME         MODE         REPLICAS   IMAGE          PORTS
8nt70z5mxdcg   bda-mysql    replicated   1/1        mysql:latest   *:3306->3306/tcp
haroon@haroon-pc:~$ docker service ps bda-mysql
ID             NAME           IMAGE          NODE           DESIRED STATE   CURRENT STATE           ERROR     PORTS
wor0w7sp33xl   bda-mysql.1    mysql:latest   Time-Machine   Running         Running 5 minutes ago
haroon@haroon-pc:~$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED          STATUS                    PORTS                                         NAMES
3c5884a3ca0d   dockersamples/visualizer   "/sbin/tini -- node …"   15 minutes ago   Up 15 minutes (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   gallant_snyde
r
haroon@haroon-pc:~$ sudo ufw allow 3306
[sudo] password for haroon:
Rule added
Rule added (v6)
haroon@haroon-pc:~$ docker service rm bda-mysql
bda-mysql
haroon@haroon-pc:~$ docker service create \
--name bda-mysql \
--constraint node.hostname==haroon-pc \
--publish mode=host,target=3306,published=3306 \
-e MYSQL_ROOT_PASSWORD=123 \
-e MYSQL_DATABASE=bda_db \
mysql
qjh5otgemdvolb8k4lvx2dwnn
overall progress: 0 out of 1 tasks
1/1: no suitable node (1 node not available for new tasks; scheduling constrain…^COperation continuing in background.
haroon@haroon-pc:~$ docker node update --availability active haroon-pc
haroon-pc
haroon@haroon-pc:~$ docker node ls
ID                            HOSTNAME       STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
i0x95xsigalwkhwn6ucrmmaec     Time-Machine   Ready     Active                          26.1.5+dfsg1
oenjz1lpyebmpn3lb5pq4nwaz *   haroon-pc      Ready     Active         Leader           28.2.2
haroon@haroon-pc:~$ docker service create --name bda-mysql --constraint node.hostname==haroon-pc --publish mode=host,target=3306,published=3306 -e MYSQL_ROOT_PASSWORD=12
3 -e MYSQL_DATABASE=bda_db mysql
Error response from daemon: rpc error: code = AlreadyExists desc = name conflicts with an existing object: service bda-mysql already exists
haroon@haroon-pc:~$ docker service create \
--name bda-mysql \
```

```
haroon@haroon-pc: ~
Error response from daemon: rpc error: code = AlreadyExists desc = name conflicts with an existing object: service bda-mysql already exists
haroon@haroon-pc:~$ docker service create \
--name bda-mysql \
--constraint node.hostname==haroon-pc \
--publish mode=host,target=3306,published=3306 \
-e MYSQL_ROOT_PASSWORD=123 \
-e MYSQL_DATABASE=bda_db \
mysql
Error response from daemon: rpc error: code = AlreadyExists desc = name conflicts with an existing object: service bda-mysql already exists
haroon@haroon-pc:~$ docker service rm bda-mysql
bda-mysql
haroon@haroon-pc:~$ docker service create \
--name bda-mysql \
--constraint node.hostname==haroon-pc \
--publish mode=host,target=3306,published=3306 \
-e MYSQL_ROOT_PASSWORD=123 \
-e MYSQL_DATABASE=bda_db \
mysql
u675ancbxyeys72tql6n6ow3a
overall progress: 1 out of 1 tasks
1/1: running
verify: Service u675ancbxyeys72tql6n6ow3a converged
haroon@haroon-pc:~$ docker service ps bda-mysql
ID             NAME          IMAGE          NODE        DESIRED STATE   CURRENT STATE            ERROR     PORTS
dampu253wpox   bda-mysql.1   mysql:latest   haroon-pc   Running         Running 26 seconds ago             *:3306->3306/tcp,*:3306->3306/tcp
haroon@haroon-pc:~$ docker ps --filter name=bda-mysql
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                                   NAMES
63e069ac7805   mysql:latest   "docker-entrypoint.s…"   12 minutes ago   Up 12 minutes   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   bda-mysql.1.dampu253wpox
i9odntojaswne
haroon@haroon-pc:~$ docker exec it 63e069ac7805 mysql --version
Error response from daemon: No such container: it
haroon@haroon-pc:~$ docker exec -it 63e069ac7805 mysql --version
mysql  Ver 9.6.0 for Linux on x86_64 (MySQL Community Server - GPL)
haroon@haroon-pc:~$ docker service rm bda-mysql
bda-mysql
haroon@haroon-pc:~$ docker service create \
  --name bda-mysql \
  --constraint node.hostname==haroon-pc \
  --publish mode=host,target=3306,published=3306 \
  -e MYSQL_ROOT_PASSWORD=123 \
  -e MYSQL_DATABASE=bda_db \
```

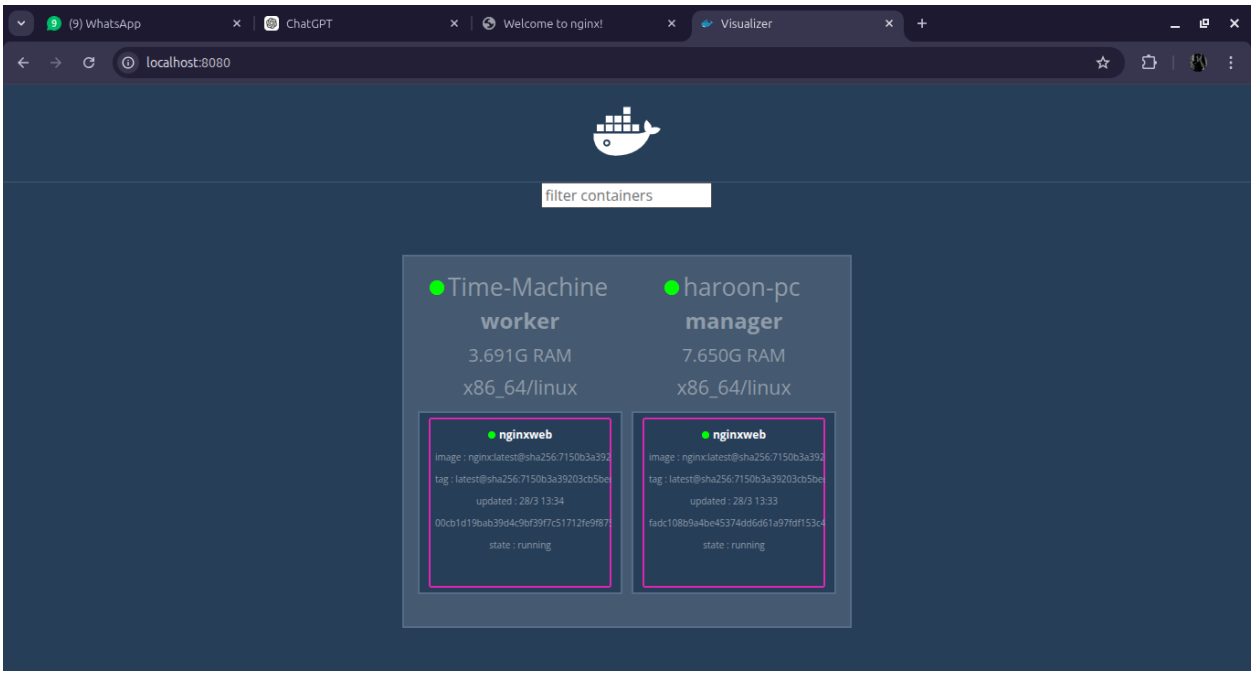
```
haroon@haroon-pc:~$ docker service create \
  --name bda-mysql \
  --constraint node.hostname==haroon-pc \
  --publish mode=host,target=3306,published=3306 \
  -e MYSQL_ROOT_PASSWORD=123 \
  -e MYSQL_DATABASE=bda_db \
  mysql:8.0
bqgcawwt6ko3924atnlw2zc0r
overall progress: 1 out of 1 tasks
1/1: running
verify: Service bqgcawwt6ko3924atnlw2zc0r converged
haroon@haroon-pc:~$ docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED          STATUS                    PORTS                                                   NA
MES
819cc2be6fec   mysql:8.0                  "docker-entrypoint.s…"   44 seconds ago   Up 43 seconds             0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   bd
a-mysql.1.72hbyon0cfpxil7y0d6cvw3od
3c5884a3ca0d   dockersamples/visualizer   "/sbin/tini -- node …"   54 minutes ago   Up 54 minutes (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp              ga
llant_snyder
haroon@haroon-pc:~$
```

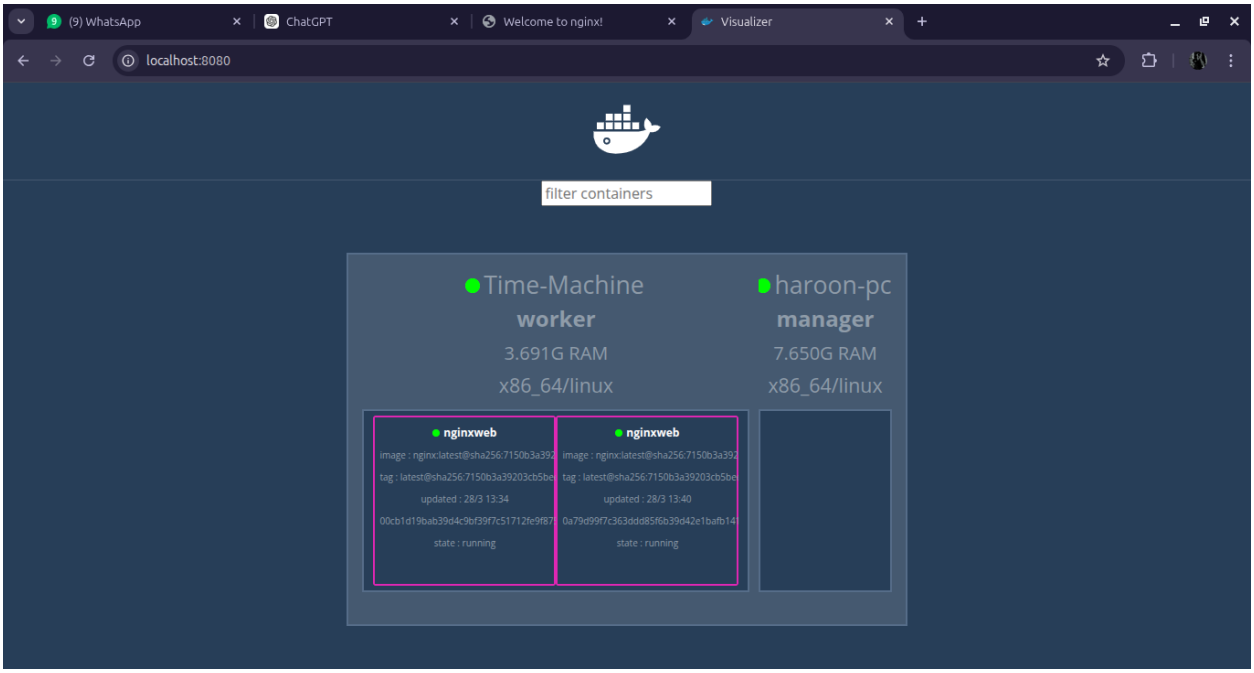
Worker Node Commands :

```
haroonpc@Time-Machine:~$ docker swarm join --token SWMTKN-1-37jlu86ka5cmyvfaofzz85rwpc7h66wd9fsgd892kut5obpt25-eo5k8zz0ws0p42tvxj7zngnd4 192.168.43.177:2377
This node joined a swarm as a worker.
haroonpc@Time-Machine:~$ ping 192.168.43.177
PING 192.168.43.177 (192.168.43.177) 56(84) bytes of data.
64 bytes from 192.168.43.177: icmp_seq=1 ttl=63 time=192 ms
64 bytes from 192.168.43.177: icmp_seq=2 ttl=63 time=215 ms
64 bytes from 192.168.43.177: icmp_seq=3 ttl=63 time=33.1 ms
64 bytes from 192.168.43.177: icmp_seq=4 ttl=63 time=54.6 ms
64 bytes from 192.168.43.177: icmp_seq=5 ttl=63 time=75.0 ms
^C
--- 192.168.43.177 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 33.147/113.962/215.053/74.674 ms
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS     NAMES
00cb1d19bab3   nginx:latest   "/docker-entrypoint.…"   50 seconds ago   Up 49 seconds   80/tcp    nginxweb.2.k2cj77fg4hc2xwe0l42w2igub
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS     NAMES
9a79d99f7c36   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   80/tcp    nginxweb.1.ti4wrgzxxw0j70rj3kc0qgval
00cb1d19bab3   nginx:latest   "/docker-entrypoint.…"   6 minutes ago        Up 6 minutes        80/tcp    nginxweb.2.k2cj77fg4hc2xwe0l42w2igub
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS                 NAMES
c93835f4bad0   mysql:latest   "docker-entrypoint.s…"   5 minutes ago   Up 5 minutes   3306/tcp, 33060/tcp   bda-mysql.1.wor0w7sp33xlxo2yw5ys7w67h
haroonpc@Time-Machine:~$ New-NetFirewallRule -DisplayName "MySQL 3306" -Direction Inbound -Protocol TCP -LocalPort 3306 -Action Allow
-bash: New-NetFirewallRule: command not found
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                 NAMES
c93835f4bad0   mysql:latest   "docker-entrypoint.s…"   13 minutes ago   Up 13 minutes   3306/tcp, 33060/tcp   bda-mysql.1.wor0w7sp33xlxo2yw5ys7w67h
haroonpc@Time-Machine:~$ docker logs c93835f4bad0
2026-03-28 08:48:14+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.6.0-1.el9 started.
2026-03-28 08:48:14+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-03-28 08:48:14+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.6.0-1.el9 started.
2026-03-28 08:48:14+00:00 [Note] [Entrypoint]: Initializing database files
2026-03-28T08:48:14.543540Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
2026-03-28T08:48:14.544714Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 9.6.0) initializing of server in progress as process 79
2026-03-28T08:48:14.551708Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-03-28T08:48:14.942757Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-03-28T08:48:16.392740Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initiali
ze-insecure option.
```

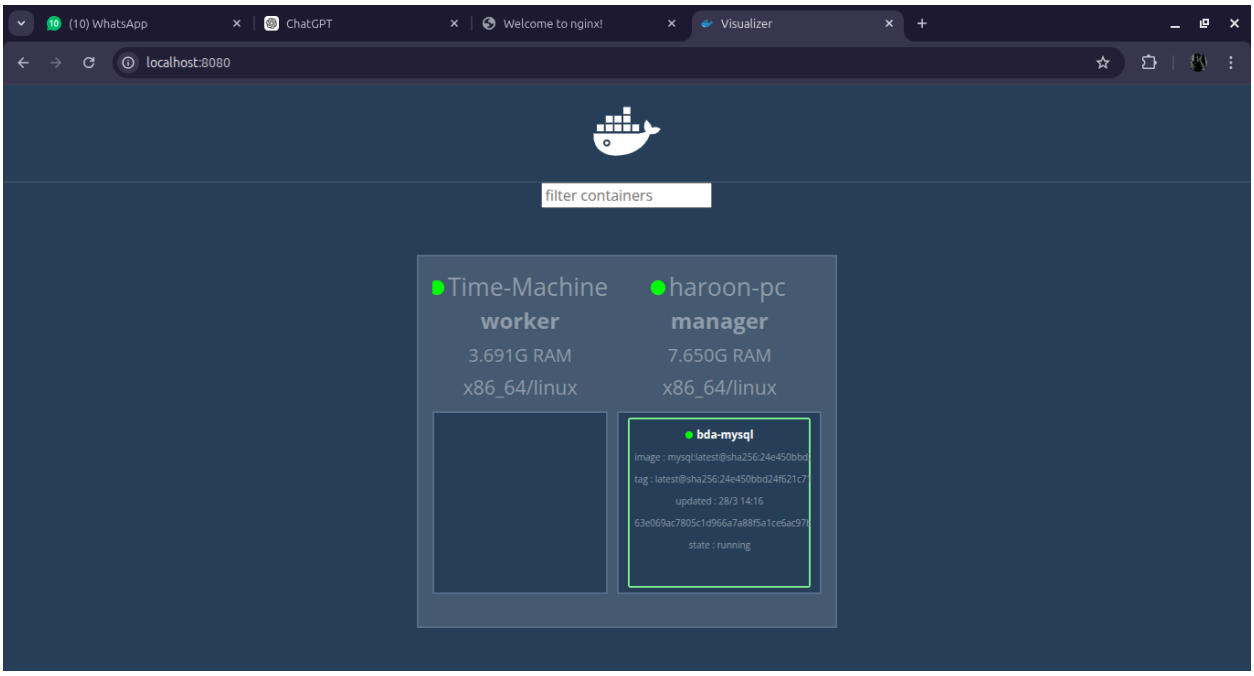
```
haroonpc@Time-Machine: ~
Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
2026-03-28 08:48:21+00:00 [Note] [Entrypoint]: Creating database bda_db

2026-03-28 08:48:21+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-03-28T08:48:21.562403Z 12 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 9.6.0).
2026-03-28T08:48:22.476603Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 9.6.0)  MySQL Community Server - GPL.
2026-03-28T08:48:22.476042Z 0 [System] [MY-015016] [Server] MySQL Server - end.
2026-03-28 08:48:22+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-03-28 08:48:22+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-03-28T08:48:22.590063Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-03-28T08:48:22.803490Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 9.6.0) starting as process 1
2026-03-28T08:48:22.803506Z 0 [System] [MY-015590] [Server] MySQL Server has access to 12 logical CPUs.
2026-03-28T08:48:22.803516Z 0 [System] [MY-015590] [Server] MySQL Server has access to 3962789888 bytes of physical memory.
2026-03-28T08:48:22.811223Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-03-28T08:48:23.222926Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-03-28T08:48:23.531274Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-03-28T08:48:23.531316Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this c
hannel.
2026-03-28T08:48:23.537490Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to
all OS users. Consider choosing a different directory.
2026-03-28T08:48:23.568612Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.s
ock
2026-03-28T08:48:23.568811Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.6.0'  socket: '/var/run/mysqld/mysqld.sock'
  port: 3306  MySQL Community Server - GPL.
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
haroonpc@Time-Machine:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
haroonpc@Time-Machine:~$
```

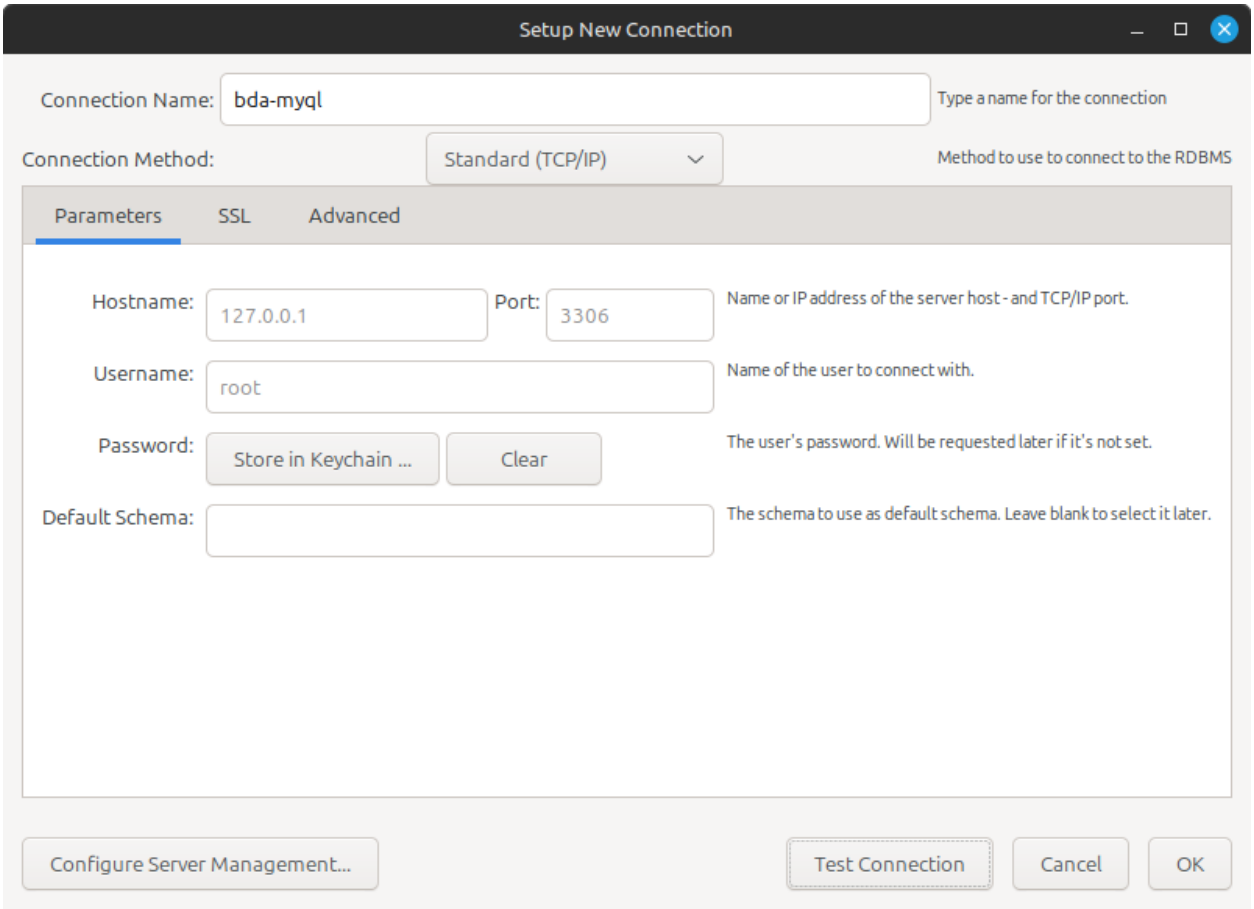
Architecture 1:

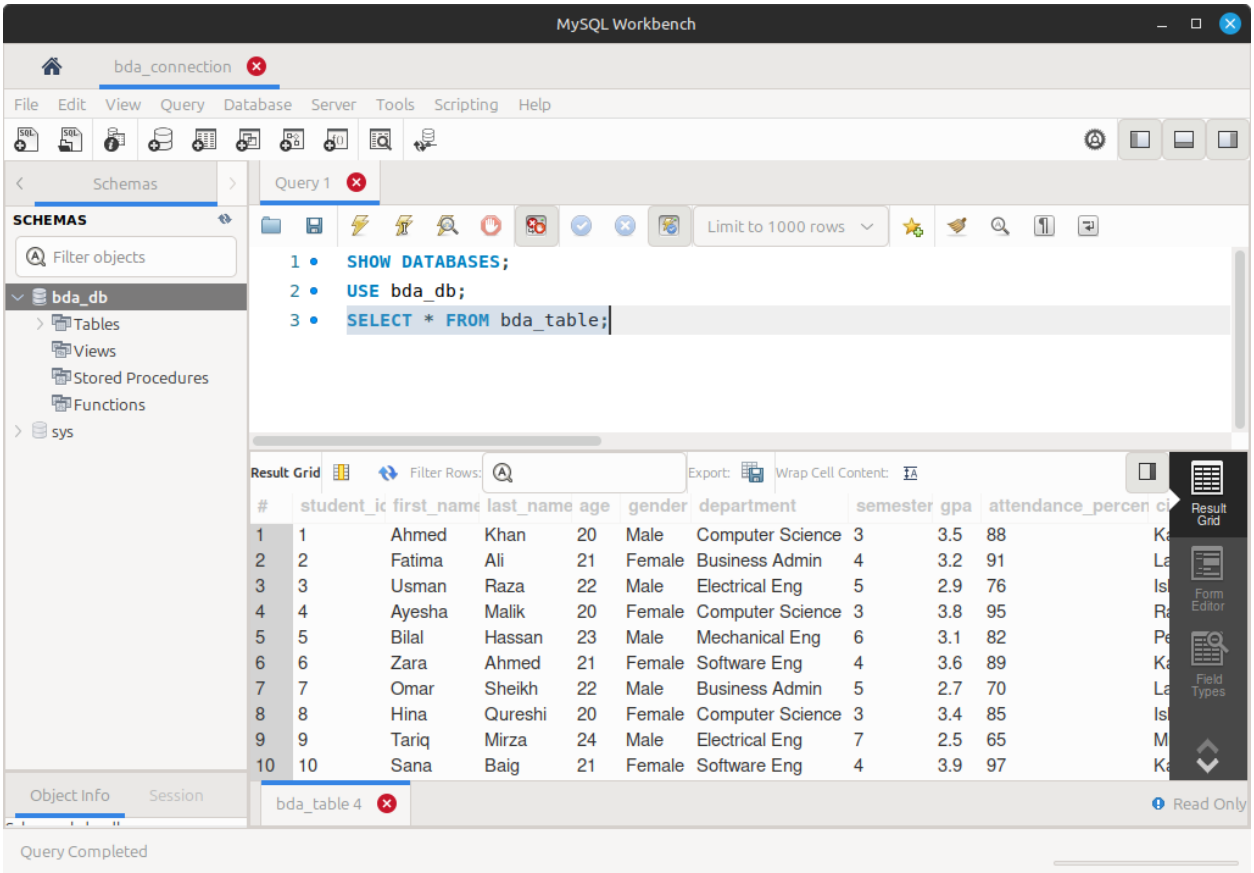




Architecture 2:




Setup New Connection
Connection Name:
bda-myql
Type a name for the connection
Connection Method:
Standard (TCP/IP)
Method to use to connect to the RDBMS
Parameters
SSL
Advanced
Hostname:
127.0.0.1
Port:
3306
Name or IP address of the server host - and TCP/IP port.
Username:
root
Name of the user to connect with.
Password:
Store in Keychain ...
Clear
The user's password. Will be requested later if it's not set.
Default Schema:
The schema to use as default schema. Leave blank to select it later.
Configure Server Management...
Test Connection
Cancel
OK



#	student_id	first_name	last_name	age	gender	department	semester	gpa	attendance_percen	ci
1	1	Ahmed	Khan	20	Male	Computer Science	3	3.5	88	Ka
2	2	Fatima	Ali	21	Female	Business Admin	4	3.2	91	La
3	3	Usman	Raza	22	Male	Electrical Eng	5	2.9	76	Isl
4	4	Ayesha	Malik	20	Female	Computer Science	3	3.8	95	Ra
5	5	Bilal	Hassan	23	Male	Mechanical Eng	6	3.1	82	Pe
6	6	Zara	Ahmed	21	Female	Software Eng	4	3.6	89	Ka
7	7	Omar	Sheikh	22	Male	Business Admin	5	2.7	70	La
8	8	Hina	Qureshi	20	Female	Computer Science	3	3.4	85	Isl
9	9	Tariq	Mirza	24	Male	Electrical Eng	7	2.5	65	Mi
10	10	Sana	Baig	21	Female	Software Eng	4	3.9	97	Ka